

A flexible modelling framework for estimating thermal tolerance and sensitivity

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Abstract

The thermal load sensitivity (TLS) and thermal death time (TDT) frameworks are gold standard approaches for estimating thermal tolerance and predict organismal vulnerability under natural thermal regimes. However, these frameworks are usually implemented through a two-step pipeline. Median times to lethal or sublethal endpoints (t_{50}) are first estimated independently across a range of assay temperatures, and these $\log_{10}(t_{50})$ estimates are then regressed against temperature. This two-step approach violates key statistical assumptions that are likely to bias the mean and variance in derived quantities such as thermal sensitivity (z), critical thermal limits (CT_{max}), heat-injury predictions, and survival probabilities.

We present a flexible Bayesian four-parameter logistic (4PL) model that estimates survival or sublethal responses as a joint function of temperature and exposure duration. We demonstrate how such an approach recovers all the classical outputs of the TLS and TDT two-stage modelling frameworks (i.e., z , CT_{max} , heat-injury integral, survival probabilities), while propagating uncertainty through all derived quantities. Using simulated data and empirical case studies in brown shrimp (*Crangon crangon*), zebrafish (*Danio rerio*), and XXX (XXX), we demonstrate the benefits of this new modelling framework.

We show that the classical and Bayesian frameworks are equivalent, but that the Bayesian approach delivers properly propagated uncertainty, can account for complex hierarchical data (i.e., random effects), can accommodate censoring and overdispersion, and unlocks outputs the two-step pipeline cannot provide, including survival landscape surfaces, between-group contrasts, and inference about the variation in the shape of thermal dose-response curves among groups. We provide a new *R* package (*bayesTLS*) and recommend the Bayesian four-parameter logistic model as the default approach for analysing TDT and TLS data, particularly when the goal is to make accurate predictions of survival under natural thermal regimes.

Introduction

Climate change is increasing the frequency, intensity, and duration of extreme heat events globally, with cascading consequences for organismal fitness, population persistence, and biodiversity (Arnold *et al.* 2025; Rezende *et al.* 2020; Smale *et al.* 2019; Snook *et al.* 2026; Wiens 2016). Quantifying variation in thermal tolerance within and among species is therefore central to forecasting biological responses to climate change in nature (Deutsch *et al.* 2008; Jørgensen *et al.* 2022; Pinsky *et al.*

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2019; Sunday *et al.* 2011, 2012). Threshold-based measures of heat tolerance, such as the critical thermal maximum (CT_{max}), have long been used to establish differences in climate vulnerability among taxa and predicting vulnerability to climate change (Deutsch *et al.* 2008; Pinsky *et al.* 2019; Pottier *et al.* 2025; Sunday *et al.* 2011, 2012). While useful, threshold-based metrics do not capture how the biological effects of extreme heat on fitness depend on both the duration and intensity of heat stress, limiting our ability to make realistic predictions of the impacts of climate change in nature (Jørgensen *et al.* 2021; Ørsted *et al.* 2022; Rezende *et al.* 2014).

The Thermal Death Time (TDT) / Thermal Load Sensitivity (TLS) models address this limitation by quantifying how the intensity and duration of heat exposure jointly determine survival responses to heat stress (Arnold *et al.* 2025; Jørgensen *et al.* 2019, 2022; Ørsted *et al.* 2022; Rezende *et al.* 2014). Recently, the TDT framework was expanded to include sublethal (e.g., fertility, heat coma) responses to heat stress, providing a flexible framework to broadly explain variation in biological responses to heat stress (Arnold *et al.* 2025; Noble *et al.* 2026; Ørsted *et al.* 2024). These models capture how heat injury accumulates in a dose-dependent manner above a certain threshold (T_{crit}), with injury accumulating more rapidly at higher temperatures and over longer exposures. As a result, organisms may eventually reach a thermal load that leads to compromised physiological function that negatively impacts growth, reproduction and survival (Arnold *et al.* 2025; Jørgensen *et al.* 2019; Ørsted *et al.* 2022; Rezende *et al.* 2014) if injury cannot be repaired (Arnold *et al.* 2025; Buckley *et al.* 2025; Jørgensen *et al.* 2021). By linking biological responses to the intensity and duration of heat stress, the TLS and TDT frameworks allow for more realistic predictions of the impacts of fluctuating thermal regimes (Jørgensen *et al.* 2021; Ørsted *et al.* 2022; Rezende *et al.* 2014), and have shown promise in predicting observed mortality rates in nature (Rezende *et al.* 2020; ?).

Despite their predictive power, TDT/TLS analyses traditionally involves a two-step modelling approach (Jørgensen *et al.* 2019; Li *et al.* 2023; Ørsted *et al.* 2022; Rezende *et al.* 2020; Truebano *et al.* 2018). First, organisms are exposed to multiple temperatures for different durations, and the proportion of individuals that are dead, infertile, knocked down, or otherwise affected is modelled as a function of exposure duration. This is typically done by fitting independent dose-response curves at each assay temperature. From these curves, the exposure duration required to reach a specified response threshold is estimated, most often t_{50} , the time required to reach 50% mortality, infertility, knockdown, or other endpoints. Second, the resulting $\log_{10}(t_{50})$ estimates are regressed against assay temperature. Thermal sensitivity (i.e., the inverse of the slope of this regression, z) and static critical thermal maximum (i.e., the intercept of the regression, $sCT_{max_{time}}$) are then derived from this linear model (Ørsted *et al.* 2022; Rezende *et al.* 2014). Thermal sensitivity (z) is interpreted as the temperature increase required to reduce t_{50} by a factor of 10, while $sCT_{max_{time}}$ is usually arbitrarily computed for a 1 min (Rezende *et al.* 2014), 1 hour (Jørgensen *et al.* 2019), or 4 hour (Ørsted *et al.* 2024) duration of heat stress exposure. These quantities can then be used to estimate cumulative Heat Injury (HI) under realistic temperature-time series, where injury accumulates as a function of the exposure duration relative to the predicted tolerance time at each temperature (Jørgensen *et al.* 2019; Villeneuve & White 2024). Recent extensions have emphasized that injury accumulation may also be partly offset by repair or acclimation mechanisms, motivating the emergence of new models that incorporate damage-repair dynamics (Arnold *et al.* 2025; Buckley *et al.* 2025).

The two-step pipeline is intuitive and easy to implement, yet it has several practical and statistical limitations. First, the two-step pipeline does not typically allow proper uncertainty propagation. The t_{50} estimates derived in the first step have non-negligible sampling variance, but they are com-

monly treated as fixed point estimates in the second-step regression. Neglecting sampling variance can bias or miscalibrate inference on (z) and (CT_{max}), and lead to inflated R^2 values - a known issue in the statistical literature (Burke *et al.* 2017; Carroll *et al.* 2006). Second, estimating t_{50} from dose-response models requires empirical data spanning a large mortality range (preferably ranging from 0 to 100% mortality) at every assayed temperature to anchor the upper and lower asymptotes of the dose-response curve. When data are sparse or do not span the full mortality range, t_{50} estimates can be incorrectly or impossible to estimate, which is then propagated to bias and miscalibrate downstream estimates of z and CT_{max} . Third, group comparisons (between species, sexes, or life stages) inherit this miscalibrated uncertainty. Reducing each dose-response curve to a fixed t_{50} estimate makes it difficult to determine if differences in thermal tolerance and sensitivity between groups can be attributed to differences in the upper asymptote, lower asymptote, or curve steepness of the dose-response curve. Fourth, fitting separate dose-response models for each assay temperature and subsequently regressing t_{50} estimates against temperature assumes that these estimates are independent, and typically ignores the hierarchical structure of experimental data (e.g., individuals nested within vials, plates, tanks, clutches, batches, populations, or experimental days). Ignoring this nesting structure prevents the decomposition of variance among nesting groups, and can lead to pseudoreplication and anticonservative uncertainty estimates, particularly when replicate-level variation affects the shape or position of the dose-response curve (Harrison 2014; Harrison *et al.* 2018; Nakagawa & Santos 2012). Finally, the two-step pipeline does not allow downstream predictions of heat injury and survival under fluctuating thermal regimes with uncertainty, which is key to inferring the confidence in our predictions of the impacts of extreme heat impacts.

Here, we present a single-stage alternative to the classical two-step pipeline: a hierarchical Bayesian four-parameter logistic model fitted jointly to all temperature-by-duration response data. In this framework, survival or sublethal response probabilities are estimated directly as a function of exposure duration and assay temperature, while allowing the parameters of the dose-response curve to vary among groups and hierarchical levels. The four-parameter logistic model provides a flexible sigmoidal description of thermal dose responses, with interpretable parameters for the lower asymptote, upper asymptote, curve steepness, and midpoint (Nielsen *et al.* 2004; Ritz 2010; Ritz *et al.* 2015; Ritz & Streibig 2008; Rudemo *et al.* 1989). We focus on binomial and beta-binomial outcomes because count data dominate TDT/TLS studies, but the same modelling logic can be extended to other response types using appropriate likelihoods, including continuous performance traits, or latency data. Similarly, although we use the four-parameter log-logistic form for its flexibility, simpler nested models (3PL, with one asymptote fixed; 2PL, with both fixed at 0 and 1) or alternative monotonic dose-response functions (e.g., Weibull, log-normal) can be implemented within the same general framework (Ritz *et al.* 2015; Ritz & Streibig 2008).

We show how classical TLS quantities, such as thermal sensitivity (z), thermal tolerance (CT_{max}), heat injury accumulation, survival under fluctuating thermal regimes, and their uncertainty, can be derived from this single Bayesian model. We implement these tools in the R package *bayesTLS* and recommend this new modelling approach as the default approach for analysing organismal responses to varying thermal loads. Using simulations and empirical case studies in brown shrimp (*Crangon crangon*), zebrafish (*Danio rerio*), and XXX (XXX), we demonstrate how to implement, validate, and expand this new modelling approach.

A non-linear Bayesian hierarchical model for characterizing thermal load sensitivity

Model specification

Assuming we have mortality data for different times and temperatures, let y denote the number of survivors out of n individuals exposed for time t (e.g., minutes) at assay temperature T ($^{\circ}\text{C}$). We can model the counts directly with a binomial likelihood (but a beta-binomial likelihood function can also be used and controls for overdispersion):

$$y \mid n, p \sim \text{Binomial}(n, p), \quad (1)$$

with survival probability p given by a four-parameter logistic (4PL) function of $\log_{10}$ -exposure time,

$$p = \text{low} + \frac{\text{up} - \text{low}}{1 + \exp(k(\log_{10} t - \text{mid}))}. \quad (2)$$

where low is the lower survival probability asymptote, up is the upper survival probability asymptote, and k is the slope defining the change in survival probability on the $\log_{10} t$ axis. Larger k gives a sharper decline in survival probability with $\log_{10} t$. mid is the value of $\log_{10} t$ at which p falls to the midpoint between low and up. When asymptotes are near 0 and 1, mid coincides with $\log_{10} t_{50}$. While other logistic forms can be used (e.g., two or 3 parameter logistics), Equation 2 is the most flexible model that can account for different asymptotic survival probabilities.

The key parameters of Equation 2 (i.e., low, up, k , mid), can be fit as a non-linear function within a likelihood or Bayesian framework, which estimates their correlation and allows for fixed or random effects to be estimated to explain variation in these parameters. Such an approach allows for *a priori* comparisons (e.g., do sexes or species differ in midpoints?), with correct inference (e.g., do species differ significantly from each other?) and allows for variance decomposition to help identify key sources of variance in the data (e.g., estimating variances explained across individuals, species, trials etc.). Such models are most flexibly fit within a Bayesian framework with intuitive interpretation of these parameters and a highly flexible path for error propagation. We show below how such a model can be used to derive all thermal load sensitivity parameters of interest to ecophysiologicalists.

Deriving key thermal load sensitivity parameters

The TDT/TLS frameworks are elegant in that they derive metrics with clear biological interpretation that can be compared across populations and species (e.g., z , $CT_{max_{1hr}}$). Here, we show that we can derive these same metrics directly from Equation 2.

As indicated above, we can model changes in the mean of any of the four parameters in Equation 2. Thermal sensitivity (z) and $CT_{max_{1hr}}$ can be derived by modelling the effect of temperature on midpoints specifically as follows:

$$\text{mid} = \beta_0 + \beta_1 (T - \bar{T}), \quad (3)$$

where mid is the $\log_{10}$ -time at which survival equals the midpoint between the lower and upper asymptotes at assay temperature T ($^{\circ}\text{C}$); β_0 is the intercept giving the midpoint at the grand mean

temperature $\bar{T}$; β_1 is the slope describing how the midpoint changes per °C increase in T ; and $\bar{T}$ is the grand mean of assay temperatures used to mean-centre T , which decorrelates β_0 from β_1 and improves parameter estimation and interpretation (Schielzeth 2010).

When the experimental design estimates upper and lower limits of the response (i.e., low and up) near 0 and 1, the midpoint coincides with the classical t_{50} (time to reach 50% survival). Therefore, the regression in Equation 3 directly provides thermal sensitivity as $z = -1/\beta_1$, and the critical thermal limit at any reference time t_{ref} is $CT_{max}(t_{\text{ref}}) = \bar{T} + (\log_{10} t_{\text{ref}} - \beta_0)/\beta_1$ (with $t_{\text{ref}} = 60$ min for $CT_{max_{1hr}}$).

The slope β_1 — and hence z — is robust to changes in asymptotes. However, threshold summaries, such as CT_{max} and $\log_{10} t_{50}$, are not. When partial mortality or sparse data do not estimate the asymptotes to be around 0 and 1, an asymmetry correction must be added to estimate these threshold summaries when using survival thresholds. For $\log_{10} t_{50}$ this involves adding the following correction factor (Equation 4):

$$\log_{10} t_{50}(T) = \text{mid}(T) + \frac{1}{k} \log \frac{\text{up}-0.5}{0.5-\text{low}}, \quad (4)$$

For CT_{max} , the correction factor must be subtracted, after division by the slope (to convert the vertical shift in $\log_{10} t_{50}$ into the corresponding shift in temperature), from the naive CT_{max} (Equation 5):

$$CT_{max}(t_{\text{ref}}) = \bar{T} + \frac{\log_{10} t_{\text{ref}} - \beta_0}{\beta_1} - \frac{1}{\beta_1} \cdot \frac{1}{k} \log \frac{\text{up}-0.5}{0.5-\text{low}}. \quad (5)$$

While we have focused on t_{50} it is important to recognise that any specific survival probability could be used outside the 50% probability threshold. For example, t_{10} marks the onset of early population mortality – useful for risk management – whereas t_{90} is closer to population-collapse. Luckily, any survival threshold can be modelled using the same posterior by adjusting the threshold in an asymmetry correction that is added to mid (Equation 6):

$$\log_{10} t_x(T) = \text{mid}(T) - \frac{1}{k} \ln \frac{\text{up} - x/100}{x/100 - \text{low}}, \quad (6)$$

CT_{max} is then adjusted in accordance through the correction factor in Equation 5. Importantly, there may be situations where computing absolute $\log_{10} t_x$ based on a sample reaching a given probability threshold is less ideal because of differences in baseline survival at the start of experiments. Instead, one might be interested in calculating what we refer to as ‘relative t_{50} ’, which we define as the time at which survival reaches the midpoint between the upper and lower asymptotes (i.e., $p = (\text{up} + \text{low})/2$). By construction, mid *is* relative t_{50} at every temperature, so no asymmetry correction is needed. CT_{max} is read directly from Equation 3 as $CT_{max}(t_{\text{ref}}) = \bar{T} + (\log_{10} t_{\text{ref}} - \beta_0)/\beta_1$. Note that relative and absolute CT_{max} are numerically equal only when $\text{up} = 1$ and $\text{low} = 0$.

A powerful aspect of the Bayesian modelling approach described above is that uncertainty in all of the estimated quantities can be derived. Additinally, each of parameters in the model (an therefore z and CT_{max}) can be estimated separately for different strata (e.g., sex, population, species) in a single joint model by including the relevant fixed covariates and/or random effects in the linear predictor for mid — and, if appropriate, in the linear predictors for low, up, and k (see Equation 11 below).

Predicting heat injury and survival under dynamic thermal environments

A particularly powerful aspect of TDT/TLS models are their ability to translate dynamic temperature times series' from nature to predict heat injury (HI) accumulation and thus population survival probability (Jørgensen *et al.* 2021; Li *et al.* 2023; Ørsted *et al.* 2022, 2024; Rezende *et al.* 2020). Heat injury accumulates as the time spent at a given temperature divided by the median lethal time at that temperature, integrated across the temperature trajectory $T(i)$ as follows:

$$HI(t) = 100 \int_0^t 10^{(T(i) - CT_{max1hr})/z} di, \quad (7)$$

where z is the thermal sensitivity and CT_{max1hr} is the static temperature at which 50% of the population reaches the assayed endpoint after one hour of exposure. When $HI(t) = 100\%$, the cumulative dose equals one t_{50} at the 1-hour reference, predicting 50% mortality (or 50% loss of the assayed sublethal endpoint, e.g. fertility; Ørsted *et al.* (2024)). Injury accumulates only when $T(i)$ exceeds a critical temperature T_c below which damage is repaired as fast as it accrues (Jørgensen *et al.* 2021; Ørsted *et al.* 2022). Field temperature records are typically available at hourly resolution. Equation 7 can then be evaluated by summing $100 \cdot 10^{(T_i - CT_{max1hr})/z}$ across each hourly interval whose temperature T_i exceeds T_c , following the implementations in Jørgensen *et al.* (2021), Ørsted *et al.* (2024), and Li *et al.* (2023). To propagate uncertainty, z and CT_{max1hr} , calculated from our Bayesian dose-response model can be used to calculate $HI(t)$ through time by substituting samples from the posterior distribution for each parameter into Equation 7.

A more natural prediction to make may instead be the population-level survival probability over time. This has the advantage of allowing for a direct connection to population dynamical models which typically use survival, growth and fecundity as vital rates for population-level predictions (Noble *et al.* 2026). To predict survival, we can calculate the *effective exposure time* at a chosen reference temperature T_{ref} . More effective time is accumulated at hot temperatures and less at cool ones, in proportion to the scaling factor $10^{(T(i) - T_{ref})/z}$,

$$\tau_{eff}(t) = \int_0^t 10^{(T(i) - T_{ref})/z} di, \quad (8)$$

We can then evaluate the 4PL surface once at the cumulative effective exposure to obtain the predicted population-level survival fraction at every point in the trajectory:

$$S_{pred}(t) = S(T_{ref}, \tau_{eff}(t)). \quad (9)$$

Here $S(T, \tau)$ is the survival surface fitted by the dose-response model, so $S_{pred}(t)$ is the survival probability for an organism held at *constant* T_{ref} after exposure duration $\tau_{eff}(t)$.

Application of Bayesian dose-response models to case studies

We applied the 4-parameter (4PL) Bayesian logistic model to four case studies covering a broad range of ectothermic groups. For all case studies, we show how much a modelling framework can retrieve classical parameters of the TLS framework such as z and CT_{max1h} using both lethal and

sublethal data. We also show how to derive an estimate of T_{crit} from sublethal data (e.g., time to heat knockdown).

In brown shrimp (*Crangon crangon*), we first show how the 4PL model can be used to generate survival curves, thermal-death-time relationships, and thermal tolerance landscapes with complete uncertainty propagation. We also show how the 4PL can be used to predict survival under dynamic thermal regimes. We then also show how to apply this model to sublethal data (time to knockdown), based on the proportion of live and knocked down individuals across time and temperature.

In zebrafish (*Danio rerio*), we show how the 4PL can be used to estimate all parameters and tolerance landscapes across three distinct life stages: early embryos, late embryos, and larvae. We show how to define models to derive group-specific parameters and compare posterior distributions to make inference about differences in thermal sensitivity and tolerance across life stages.

In XXX...

To break the well-known likelihood ridges among $(\text{low}, \text{up}, k)$ in unconstrained 4PL fits (Bolker 2008; Gelman 2006), we estimate the identifiable parameterisation

$$\text{low} = 0.49 \logit^{-1}(\widetilde{\text{low}}), \quad \text{up} = 0.51 + 0.49 \logit^{-1}(\widetilde{\text{up}}), \quad k = \exp(\kappa), \quad (10)$$

with $\widetilde{\text{low}}, \widetilde{\text{up}}, \kappa \in \mathbb{R}$. This constrains $\text{low} \in [0, 0.49]$ and $\text{up} \in [0.51, 1]$ into disjoint intervals so that $\text{up} > \text{low}$ by construction, the curve direction is unambiguous, and k is positive on the natural scale.

Up to this point we have written Equation 1–Equation 3 without indices for clarity. To accommodate replicated, hierarchically structured data, we now index observations by i within group j (e.g., trial, individual, population). The inflection point mid_{ij} is then modelled as a linear function of (mean-centred) assay temperature, with additional fixed covariates and nested random effects:

$$\text{mid}_{ij} = \beta_0 + \beta_1 (T_{ij} - \bar{T}) + \mathbf{x}_{ij}^\top \gamma + b_j + \varepsilon_{ij}, \quad (11)$$

where $\bar{T}$ is the grand mean of assay temperatures (centring stabilises sampling and decorrelates β_0 from β_1), $\mathbf{x}_{ij}$ holds any additional fixed covariates with coefficients γ , b_j collects the nested random effects and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is an observation-level random effect that absorbs the binomial overdispersion characteristic of replicated count assays (Harrison 2014). However, we could also remove the observation-level random effect from the model and instead fit a beta-binomial error distribution which estimates a dispersion parameter (ϕ) more naturally.

Together, Equation 1–Equation 11 define the hierarchical Bayesian 4PL fit directly to the raw binomial counts. Every classical TDT quantity — slope, z , CT_{max} at any reference time, and the heat-injury integral — is recovered as a one-line posterior summary of this single joint fit, as derived in §[Equivalence with the classical pipeline].

What the 4PL delivers that the two-step cannot

Worked example: brown shrimp thermal tolerance

Data and experimental design

Lethal TDT — hierarchical 4PL on counts

Sublethal time-to-death

Dynamic CT_{max} across ramping rates

Reconciling the three endpoints via the Jørgensen ramping equation

Field-temperature heat-injury projections under warming scenarios

Discussion

What changes for the field if the hierarchical 4PL becomes the default

Limits of the present analysis

Recommendations for primary studies and meta-analyses

Conclusions

Acknowledgements

Author contributions

Data and code availability

All data, analysis scripts, and the rendered manuscript and supplement are available at <https://github.com/daniellnoble/bayesTLS>.

Conflict of interest

The authors declare no competing interests.

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